Section-13

PORCELAIN DISC INSULATORS

1.0 SCOPE : Refer Clause-1 of Section-I, General

This specification covers design, manufacture, testing at manufacturer's works before dispatch and supply of porcelain disc insulators (Normal) as per the requirements given in Schedule of Requirements.

2.0 STANDARDS:

2.1 The insulators shall comply in all respects with the Indian standard specification IS-731-1976 with latest amendments or IEC publication 383-1993, IEC 433-1980 or any other latest authoritative standard.

2.2 Insulators conforming to any other internationally accepted standards, which ensure equal or higher quality than the standard will also be acceptable. Where the material is offered according to any such international standard, an English version of the specification shall be attached to the tender.

3.0 ATMOSPHERIC CONDITIONS: Refer Clause-2 of Section-I, General

3.1 **SYSTEM CONDITIONS.**

Electrical system Data:

Electrical system Data:

a)	System Voltage (KV rms)	:	220
b)	Max. Voltage (KV rms)	:	245
c)	Lighting Impulse withstand voltage (Dry & Wet) (KVP)	:	1050
d)	Power frequency withstand voltage (wet) (KVP)	:	460
e)	Short circuit level and duration	:	40 KA per one second.

3.2 The materials offered shall be suitable for operation in tropical climate in which they will be subject to the full rays of the sun and inclement weather and shall be able to withstand wide range of temperature variations. The humidity may be as high as 100% during rainy season and as low as 10% during dry season. The prevailing weather conditions are given above.

3.3 Reference atmospheric conditions for which insulator characteristics shall be expressed for the purpose of comparison are given below.

Ambient temperature	20 Deg C
Barometric pressure	1013 Millibars
Absolute humidity	11 gms. Of water per cubic Metre corresponding to 63 percent relative humidity at 20 Deg.C

3.4 Tests for the purpose of this specification may be carried out under conditions naturally obtaining at the time of the test. The barometric pressure, air temperature and humidity shall be recorded for the purpose of corrections.

4.0 GENERAL REQUIREMENTS:

4.1 All materials used in the manufacture of the disc insulators shall be of the best quality.

4.1.1. The glaze shall cover all porcelain parts of the disc insulators except those areas, which serve as supports during firing which are left unglazed for the purpose of assembly.

4.1.2. Cement used in the construction of disc insulators shall not cause fracture by expansion or loosening by contraction and proper care shall be taken to locate the individual parts correctly during cementing. The cement shall not give rise to chemical reaction with metal fittings and its thickness shall be as uniform as possible.

4.2 DESIGN AND TYPE

4.2.1 Type B insulators shall be offered. Type B insulator is one in which the length of the shortest puncture path through solid insulating material is less than half the length of the shortest flashover path through air outside the insulator.

4.2.2 The technical particulars of the disc insulators required and the system requirements are indicated in Annexure-I & clause '3.1' respectively. The design of insulators shall be such that stresses due to expansion and contraction in any part of the insulator shall not lead to deterioration.

4.2.3 The disc insulators shall be of ball and socket type. The ball and socket of the disc insulator shall be galvanized and mechanically strong. The ball shall move free in the sockets, but shall be so designed that they do not give way while in service.

4.2.4 The sockets shall be made of malleable iron. These shall be free from cracks, shinks air holes, burrs and rough edges. The socket shall be circular with inner and outer surfaces concentric and of such design that they will not yield or distort under the stress to the porcelain shells.

4.2.5 MATERIALS:

The ball pin shall be a single piece made of drop forged steel and shall be free from laps, folds, burrs and rough edges. All bearing surfaces shall be smooth and uniform so as to distribute the loading stresses uniformly. The ball pins shall be of such a design that they will not yield or distort under loaded condition. The pins shall not be made by joining, welding, shrink fitting or any other pieces from more than one piece of material.

4.2.6 Security clips for use with ball and socket coupling shall be in accordance with IS 2486 (Part III & IV) and made of suitable material of stainless steel or copper alloy or Phosphor-bronze. The security clips shall provide positive locking of the coupling. The logs of the security clips shall be spread after installation to prevent withdrawal from the socket.

4.3 PORCELAIN:

The porcelain used in the manufacture shall be sound, free from defects, thoroughly vitrified and smoothly glazed. The porcelain shall not engage directly with the hardware.

4.4 GLAZE:

The finished porcelain shall be glazed in brown colour. The glaze shall cover all exposed porcelain parts of the insulators, shall have a good luster, smooth surface and a good performance even under extreme weather conditions of tropical climate. It shall not get cracked or chipped by ageing under normal service conditions. The glaze shall have coefficient of expansion as that of the porcelain body throughout the working temperature range.

4.5 DIMENSIONS AND TOLERANCES:

The disc insulators covered by this specification shall have the dimensions specified in Annexure-I subject to the tolerances indicated in relevant IS.

4.6 WORKMANSHIP:

4.6.1 The insulators covered by this specification shall be of latest design and conform to the latest scientific methods and shall be suitable for use on 132 KV and 220 KV transmission lines, Sound design, latest manufacturing processes and proper material quality control shall be ensured at various stages. Good finish and elimination or sharp edges and corners shall also be ensured. The disc insulator shall be free from all sorts of defects. All exposed surfaces shall be smooth and perfect.

4.6.2 The forged metal sockets and shank pins shall be free from cracks, seams, shrinks, air-holes and rough edges. Metal pins shall be free from laps, folds, seams, burrs and rough edges. All surfaces of metal parts shall be perfectly smooth with no projecting points of irregularities.

4.6.3 All the ferrous parts shall be hot dip galvanized in accordance with the latest edition of IS 2629 and shall satisfy the tests mentioned in latest edition of IS 2633-1972. The zinc coating shall be adherent, smooth, reasonably bright, continuous and free from such imperfections like flux, ash, rust stains, bulky white deposits and blisters. Bituminous paint coating shall be provided on the metal units after assembly shall be concentric and co-axial within the limits as permitted by the relevant Indian.

The zinc used for galvanizing shall be of grade Zn.99.95 as per IS-209-1979.

4.7 SIZE AND NUMBER OF DISCS:

The size and number of discs to be provided in the insulator strings is indicated in Annexure-I.

4.8 INTER CHANGEABILITY:

The insulators inclusive of the ball and socket fittings shall be of standard design suitable for use with hardware fittings of any make conforming to relevant Indian Standards.

5.0 TESTING:

5.1 Suitable number of individual discs and accessories and complete strings shall be subjected to the tests as per latest editions of IS 731-1976, IS 2486 (Part-II)-1971, IEC 383 or any other equivalent authoritative standard.

5.2 TEST VOLTAGE:

The test voltage of the insulator shall be as given in Annexure-I.

5.3 MECHANICAL LOADS:

The insulator strings shall be suitable for the minimum failing loads specified in Annexure-I. The load shall be applied axially to the insulator strings.

5.4 TESTS:

5.4.1 **TYPE TESTS:**
The type tests must have been conducted not earlier than five years form the date of bid opening. The Bidder shall furnish one set of the following type test reports as per latest editions of IS:731-1976 & IEC-383-1993 along with offer.

a) Visual examination

b) Verification of dimensions

c) Visible discharge test

d) Impulse voltage withstand test

e) Power frequency voltage withstand test
Under (I) dry and wet condition and flashover test

f) Temperature cycle test

g) Electro-Mechanical failing load test

h) Puncture test

i) Porosity test

j) Galvanizing test

5.4.3 ROUTINE TESTS:

The following shall be carried out as routine tests.

a) Visual examination

b) Mechanical routine test

c) Electrical routine test

5.4.4. ACCEPTANCE TESTS:

The test samples after having withstood the routine tests shall be subjected to the following acceptance tests as per latest edition of IS 731 in the order indicated below.

a) Verification of dimensions

b) Temperature cycle test

c) Twenty four hours mechanical strength test

d) Electro-Mechanical failing load test

e) Puncture test and

f) Porosity test and

g) Galvanizing test.

6.0 QUALITY ASSURANCE PLAN:

The Bidder shall invariably furnish following information along with his offer, failing which the offer shall be liable for rejection. Information shall be separately given for individual type of material offered.

i) Statement giving list of important raw materials, names of sub-suppliers for the raw material, list of standards according to which the raw materials are tested, list of tests normally carried out on raw materials in the presence of Bidders representative and copies of test certificates.

ii) Information and copies of test certificates as in (i) above in respect of bought out items.

iii) List of manufacturing facilities available.

7.0 MARKING:

7.1 Each insulator shall be legibly and indelibly marked to show the following:

a) Name or trade mark of the manufacturer

b) Month and year of manufacture

c) Minimum failing load in Kg or KN

d) Country of manufacture.

7.2 Markings on porcelain shall be printed and shall be applied before filing. The words **"APTRANSCO"** must be etched on the porcelain.

8.0 CREEPAGE DISTANCES:

The tenderer shall specify in the tender the creepage distance of the insulators. The minimum creepage distance required for insulators, to be used with their axis approximately vertical are given in Annexure – I.

9.0 INSPECTION:

The purchaser's representative shall have access to the manufacturer's works for purpose of inspection of the manufacture of the materials covered by this specification. 15 days notice shall invariably be given to the APTRANSCO for arranging inspection of the materials. The APTRANSCO will not be responsible for any delay if notice of 15 days is not given.

10.0 PACKING:

All insulators shall be packed in strong seasoned wooden crates suitably for easy but rough handling acceptable for rail and sea transport. When packed in a crate, as a string wooden separators shall be fixed between the insulators to keep individual insulator in position without movement within the crate.

11.0 GUARANTEED TECHNICAL PARTICULARS:

The number of disc insulators used on suspension and tension strings and the Electro mechanical strength are given in Annexure-I. Full guaranteed particulars including dry and wet flashovers, puncture and impulse voltages, corona formation voltages, creepage distances, length, voltage distribution on strings etc., shall be furnished.

12.0 SCHEDULE OF REQUIREMENTS AND DESIRED DELIVERY AND PRICES

12.1 The schedule of requirements and desired deliveries are indicated in Schedule of Requirements.

12.2. The schedule of prices should be indicated in the same Form

12.3 The prices shall be FIRM.

13.0 SCHEDULE OF REQUIREMENTS:

13.1 The requirement of materials indicated are only provisional and final quantities will be confirmed in the Purchase order.

13.2 The quantity mentioned in this specification is only provisional and APTRANSCO shall have the right to place orders for 25% more in addition to the quantity mentioned in the specification during the period the original order is under execution.

13.3 The purchaser reserves the right to place the order for part or full quantity or to reject any of the tenders without assigning any reasons.

14.0 DRAWINGS:

Fully dimensioned drawings shall be enclosed in duplicate to the tender, otherwise tender is liable for rejection. Drawing for purpose of tendering is enclosed.

15.0 SAMPLES:

The tenderers shall handover the samples in Room No.551'A', Vidyut Soudha, Hyderabad-82 at the time of submission of their tenders. No responsibility will be taken by this office to take delivery of the sample sent by road or rail. The tenderers themselves shall arrange for handing over the samples at this office personally and obtain proper acknowledgement. It may please be noted that the samples will be tested for proper performance and hence they cannot be returned to any tenderer. The samples shall become the property of the APTRANSCO. Tenders received without proper drawings and samples are liable to be rejected. Tenderers who have supplied insulators previously to APTRANSCO, need not supply the samples.

ANNEXURE – I

REQUIRED RATING AND ELECTRICAL CHARACTERISTICS

Sl. No.	Characteristics	Type of Insulators			
		70 KN B&S type Normal Disc Insulators.	120 KN B&S type Normal Disc Insulators	70 KN B&S type Anti-fog Disc Insulators.	120 KN B&S type Antifog Disc Insulators
1.	Type description	Ball & Socket type	Normal disc insulators	Ball & Socket type	Anti-fog disc insulators
2.	Colour & surface of porcelain portion	BROWN GLAZED			
3.	Ferrous parts	AS PER IS-2629 & 2633 with latest editions			
4.	Check clip provided In the socket.	'W' type clip with Phospher Bronze	R type clip with stainless steel.	'W' type clip with Phospher Bronze	'R' type clip with stainless steel.
5.	Ball pin	Drop Forged Steel			
6.	Socket fitting	Malleable Cast Iron/Sperodical Graphite Iron			
7.	Ball pin designation	16 'B' mm	20mm	16 'B'mm	20mm
8. a)	Diameter and spacing of insulators (in mm).				
	Diameter	255	280	255	280
	Spacing	145+4	145+4	145+4	145+4
b)	Tolerances on diameter and spacing				
	(i) Dia +/-	(0.04 d + 1.5) mm where 'd' is the dia of shell d≤ 300m			
	(ii) spacing +/-	(0.025 S + 6) mm where 'S' is the spacing of disc d > 300m			
9.	Minimum failing Load of the insulator (Kg)	7000	11500	7000	11500
10.	Minimum total creepage distance in mm.	295	320	435	435
11.	Minimum protected creepage distance in mm	175	215	240	240
12.	Combined Mechanical and Electrical strength in KGs.	7000	11500	7000	11500
13.	One minute power Frequency withstand voltage (kV) Dry	75	75	80	80
14.	One minute power Frequency withstand voltage (kV) Wet	48	48	48	48
15.	Flashover Power Frequency voltage (kV)				
	Dry	80	80	85	85
	Wet	45	45	50	50
16.	Visible discharge voltage (kV)	9	18	9	18
17.	Puncture Voltage (kV)	120	120	140	140

Contd…..

Sl. No.	Characteristics	Type of Insulators			
		70 KN B&S type Normal Disc Insulators.	120 KN B&S type Normal Disc Insulators	70 KN B&S type Anti-fog Disc Insulators.	120 KN B&S type Antifog Disc Insulators
18.	Impulse flash over voltage (kV)				
	Positive	125	125	125	125
	Negative	130	130	130	130
19.	Standards according to which the porcelain insulator shall be manufactured and tested.	IS-731-1971 with latest amendments.			
20.	No. of units for suspension/tension String				
	132 KV	9	10	9	10
	220 KV	13	14	13	14
21.	Rated Voltage of each insulator (kV)	12	12	12	12

Annexure - II

GUARANTEED PERFORMANCE & OTHER PARTICULARS

(While furnishing these details, the values in Annexure-I also are to be furnished.)

Sl. No.	Description	Electro Mechanical Strength			
		For Normal & Antifog type strings			
		70 KN (Normal)	120 KN (Normal)	70 KN (Antifog)	120 KN (Antifog)
(1)	(2)	(3)	(4)	(5)	(6)
1.	a) Type of insulator (Ball and socket)				
	b) Size and designation ball and socket with standard to which it will conform.				
2.	Maker's name, address and country.				
3.	Dimensions:				
	a) Porcelain Disc diameter in mm				
	b) Unit spacing in mm.				
	c) Creepage distance in mm.				
	i) Total				
	ii) Protected				
4.	Mechanical Values:				
	a) Combined Mechanical and Electrical strength in KGs.				
	b) Mechanical breaking strength in KGs.				
	c) Routine Mechanical test load (KGs.)				

(1)	(2)	(3)	(4)	(5)	(6)
5.	Electrical Values:				
	a) Power frequency Flashover Voltage DRY (KV)				
	b) Power frequency Flashover Voltage WET (KV)				
	c) Impulse flashover voltage 1.2 x 40 mircro second negative wave (KV)				
	d) Impulse flashover voltage 1.2x40 mircro Second negative wave (KV)				
	e) Power frequency withstand voltage : DRY (KV)				
	f) Power frequency withstand voltage : WET (KV)				
	g) Impulse withstand test voltage 1.2x40 micro second positive and negative wave (KV)				
	h) Power frequency puncture Voltage (KV)				
6.	Net weight of insulator unit in KGs.				
7.	Strandard specification to which insulators will conform.				
8.	Other particulars if any.				

SIGNATURE OF THE TENDERER